paper.

CUTROOM-DECK EXAMPLE

Fuzzy search over municipality names

How often Levenshtein distance finds the right town despite typos

5,571 IBGE municipalities and 22,284 queries measured in October 2026

SUMMARY

With up to 3 typos, the search ranks the right municipality first in 85.5% of queries

VERDICT

Good for suggesting, not for correcting on its own

The right name is first in 85.5% of queries with 1 to 3 errors and in the top 5 in 94.9%. Exact text matching gets 0.4%.

RIGHT NAME FIRST

85.5%

exact match: 0.4%

IN THE TOP 5

94.9%

in a list of 5 suggestions

WITH 1 ERROR

95.4%

right name first

Removing accents before comparing fixes accent errors (5.0% → 0.1% misses), but the total drops to 85.0% from 85.5%.

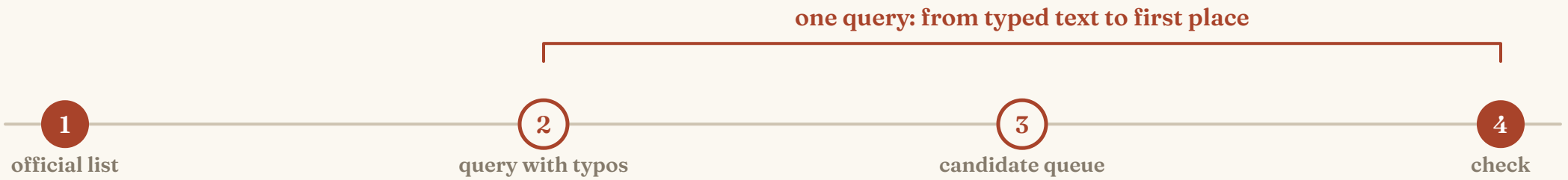
Swapping two neighboring letters pushes the right name out of first place in 12.7% of 1-error queries.

Names of up to 6 letters are 14% of queries and 40% of the misses.

IBGE list of municipalities (localities API), downloaded Oct 21, 2026: 5,571 municipalities, 5,298 distinct names. 16,713 queries with 1 to 3 errors, lowercase text.

HOW WE MEASURED

We built queries with typos from the official list and checked where the right name lands in the queue



1

Official list

5,571 IBGE municipalities, 5,298 distinct names.

2

Queries with typos

Each name gets 0, 1, 2 and 3 random errors (accent, missing, extra, wrong or swapped letter): 22,284 queries.

3

Candidate queue

The 5,298 names are ranked by distance to the query, nearest first.

4

Check

Is the right name first? In the top 5? The next pages measure that.

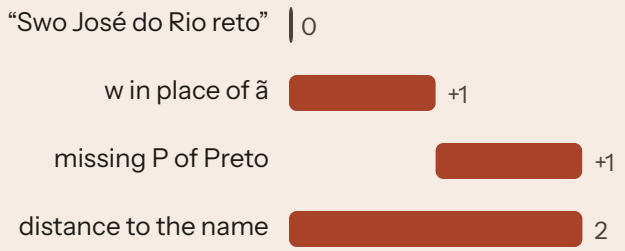
IBGE list of municipalities (localities API), downloaded Oct 21, 2026: 5,571 municipalities, 5,298 distinct names. Errors drawn with a fixed seed (2026), so the result is reproducible.

WHAT DISTANCE IS

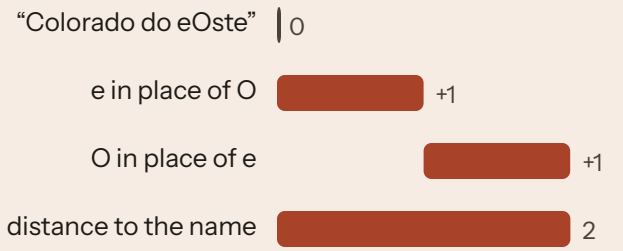
Distance counts the edits from the query to the name, and the search ranks names by it

Levenshtein distance adds up the one-letter edits (replace, remove or insert) needed to turn the typed text into the name. Lower is closer. The search puts the name with the lowest distance first.

QUERY WITH 2 ERRORS → SÃO JOSÉ DO RIO PRETO



QUERY WITH 1 ERROR (SWAPPED LETTERS) → COLORADO DO OESTE



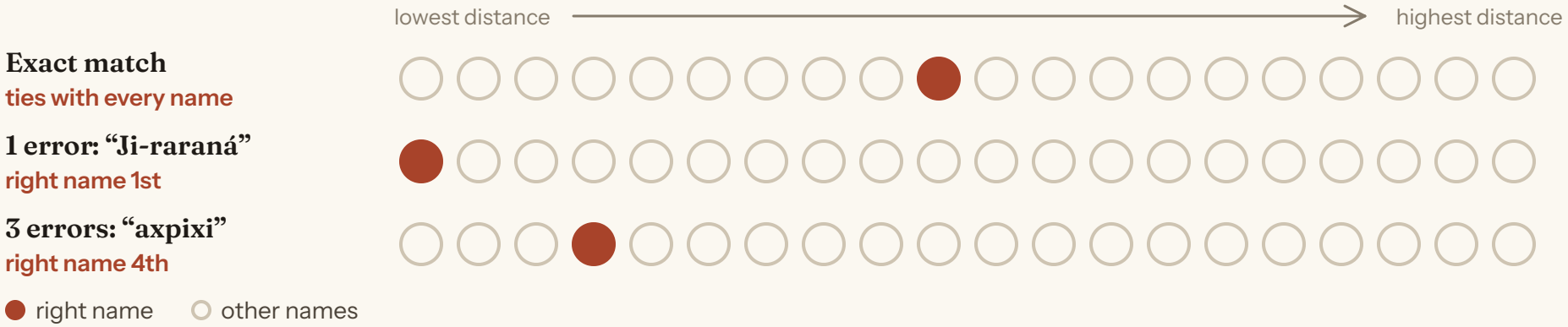
HOW TO READ

Swapping two neighboring letters is a single slip for whoever types, but Levenshtein distance counts two edits. That gap shows up again in the results.

Real examples from the query set; distances computed on lowercase text.

WHAT RANKING IS

For whoever searches, what matters is the right name being first in the candidate queue



The first 20 candidates of each query. For real queries the position is the measure; for exact matching the right name ties with all of them and the position shown is illustrative.

HOW TO READ

- Each queue has all 5,298 names, from nearest to farthest.
- **Ranking (AUC)** measures the whole queue: on average 1.4 wrong name lands ahead of the right one, out of 5,297.
- **Right name first** measures only the front of the queue, which is what people see.

IN THE 3-ERROR QUERY (CABIXI)

“axpixi” is 2 edits from Axixá and 3 from Apiaí, Apodi and Cabixi: the right name ties with two others and falls to 4th.

Real queries from the set; ties at the same distance broken alphabetically in this illustration only.

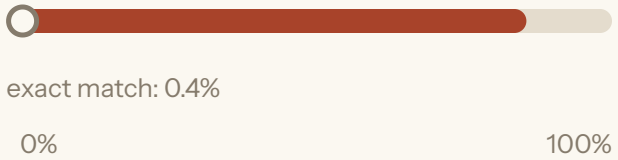
THE THREE MEASURES

Three numbers sum up the search: first place, the top 5, and the ranking of the whole queue

RIGHT NAME FIRST

85.5%

Out of 100 queries with typos, how many have the right name first.

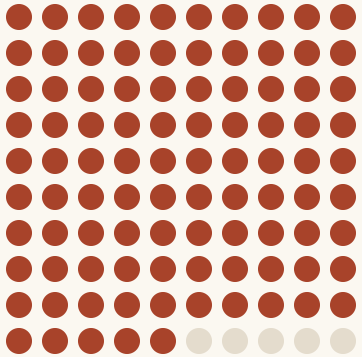


Higher is better.

IN THE TOP 5

95 of 100

How many show the right name in a list of 5 suggestions.



Higher is better.

RANKING · AUC

0.9997

Between a wrong name and the right one, the chance the right one comes first.

1.4

wrong name ahead of the right one, on average, out of 5,297

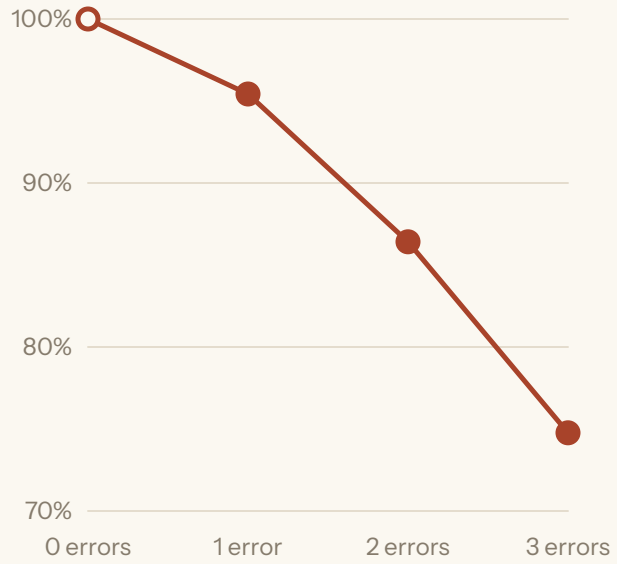
Closer to 1 is better. 0.5 is a coin flip.

IBGE list of municipalities (localities API), downloaded Oct 21, 2026: 5,571 municipalities, 5,298 distinct names. 16,713 queries with 1 to 3 errors, lowercase text.

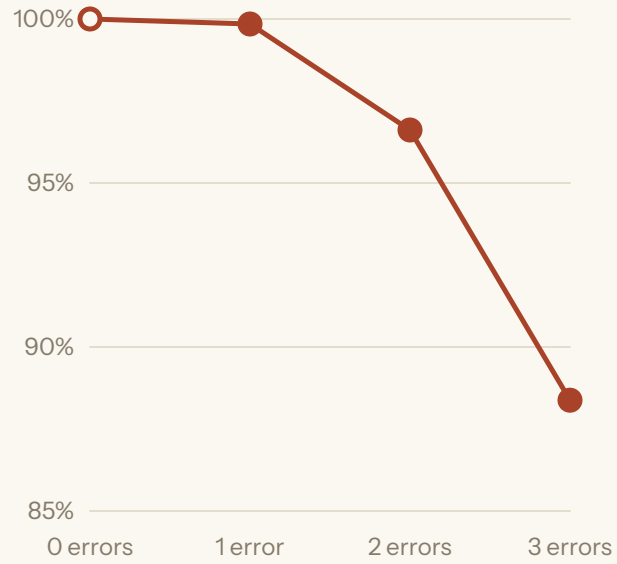
TYPING ERRORS

Each extra error lowers first-place hits: from 95.4% with 1 error to 74.8% with 3

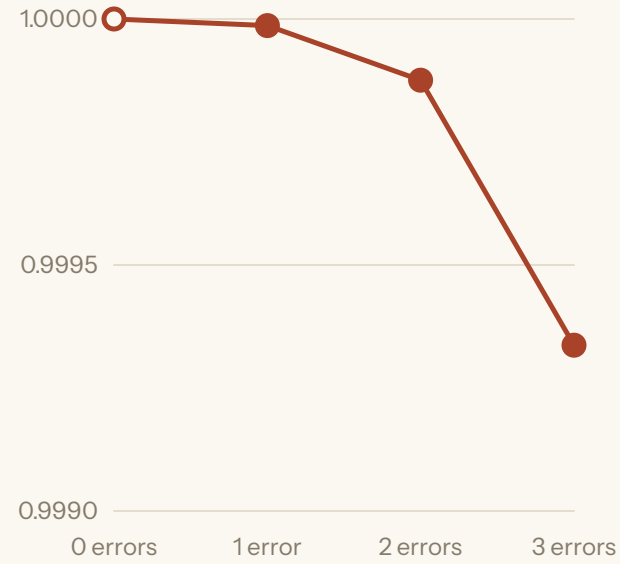
Right name first



In the top 5



Ranking · AUC



With 3 errors, the right name is still in the top 5 for 88.4% of queries. Hollow points: queries without errors, only as a reference.

IBGE list of municipalities (localities API), downloaded Oct 21, 2026: 5,571 municipalities, 5,298 distinct names. 5,571 queries per number of errors, lowercase text.

WHICH NAMES

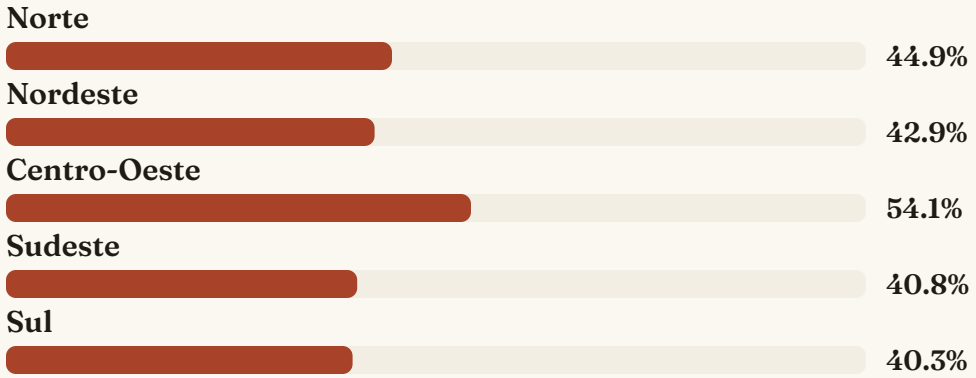
43% of the names have an accent and 48% have more than one word

How municipalities split by name shape



One word, no accent	32.3%
One word, with accent	19.9%
Compound, no accent	24.9%
Compound, with accent	22.9%

% of names with an accent, by region



HOW TO READ

Accent, length and region are the cuts of the next pages. 9.1% of municipalities share their name with another municipality in another state: by name alone, the search cannot tell them apart.

IBGE list of municipalities (localities API), downloaded Oct 21, 2026: 5,571 municipalities, 5,298 distinct names.

BY REGION

First-place hits go from 84.2% in the Sudeste to 87.7% in the Sul

Region	% of municipalities	Right name first · 95% CI	Accents removed	Queries
Norte 450 municipalities · names of 12.2 letters on average	8.1%	 85.5% (83.5% to 87.3%)	85.1%	1,350
Nordeste 1,794 municipalities · names of 11.5 letters on average	32.2%	 85.2% (84.2% to 86.1%)	84.3%	5,382
Centro-Oeste 468 municipalities · names of 11.9 letters on average	8.4%	 86.1% (84.2% to 87.8%)	86.5%	1,404
Sudeste 1,668 municipalities · names of 11.4 letters on average	29.9%	 84.2% (83.2% to 85.2%)	84.0%	5,004
Sul 1,191 municipalities · names of 11.7 letters on average	21.4%	 87.7% (86.6% to 88.8%)	87.0%	3,573

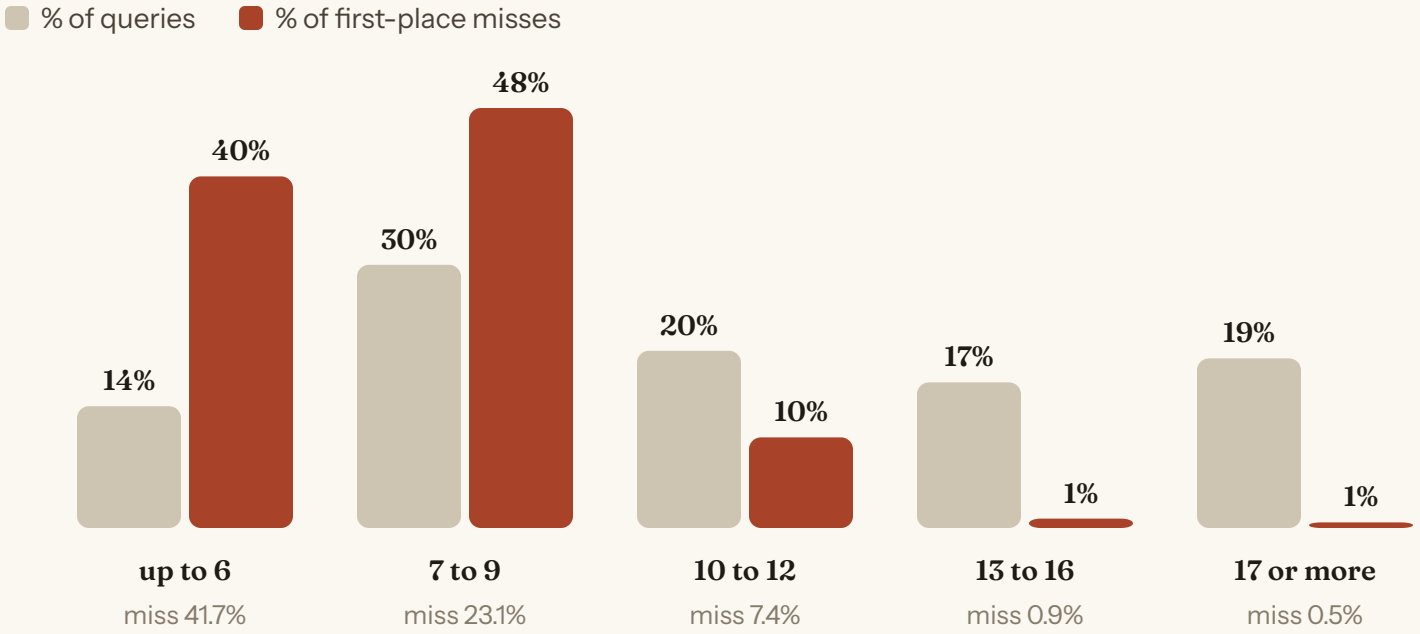
HOW TO READ

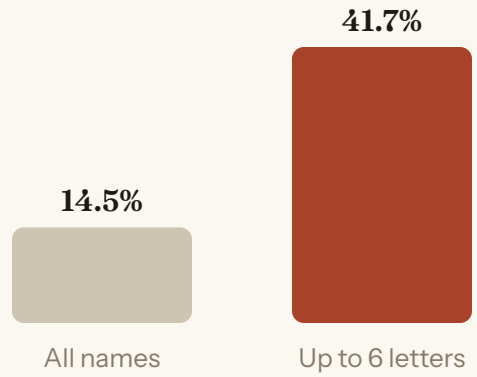
The dot is the first-place rate; the line, its 95% confidence interval. Intervals that do not overlap, like the Sudeste's and the Sul's, point to a difference beyond chance. Mean name length barely varies across regions and does not follow the rate; the cause of the difference was not investigated.

IBGE list of municipalities (localities API), downloaded Oct 21, 2026: 5,571 municipalities, 5,298 distinct names. 16,713 queries with 1 to 3 errors, lowercase text.

NAME LENGTH

Names of up to 6 letters are 14% of queries and 40% of the misses





HOW TO READ

A colored bar taller than the grey one means the band concentrates misses. For names of up to 6 letters the miss rate is 41.7%, 2.9x the average: each error weighs more in a short name, and more names sit a few edits away.

IBGE list of municipalities (localities API), downloaded Oct 21, 2026: 5,571 municipalities, 5,298 distinct names. 16,713 queries with 1 to 3 errors, lowercase text. Length in characters, spaces included.

LENGTH × ERRORS

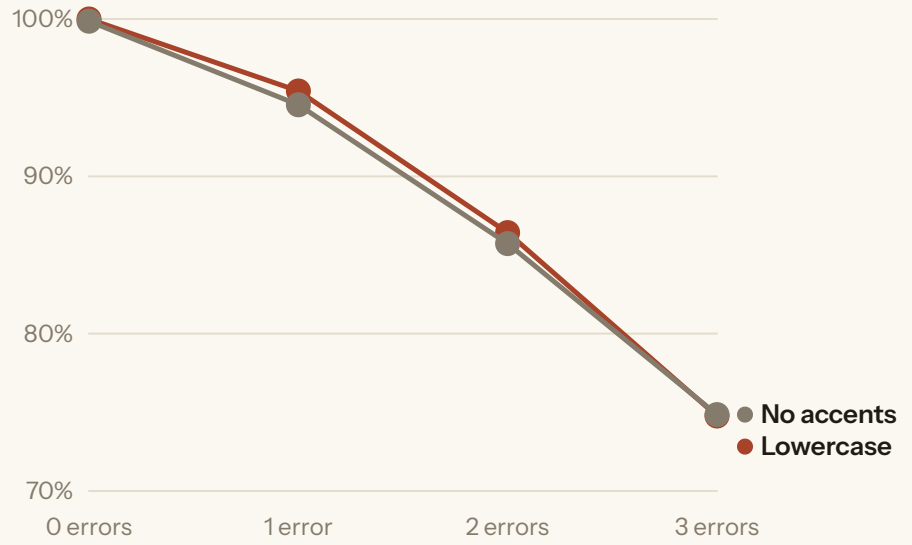
With 3 errors, names of up to 6 letters come first in 35% of queries

% of queries with the right name first, by name length (letters)

	1 error	2 errors	3 errors
up to 6	82.9%	56.8%	35.1%
7 to 9	94.2%	78.8%	57.7%
10 to 12	97.9%	94.8%	85.1%
13 to 16	99.9%	99.0%	98.3%
17 or more	99.9%	99.9%	98.8%

Stronger color: more misses.

Right name first: lowercase × accents removed

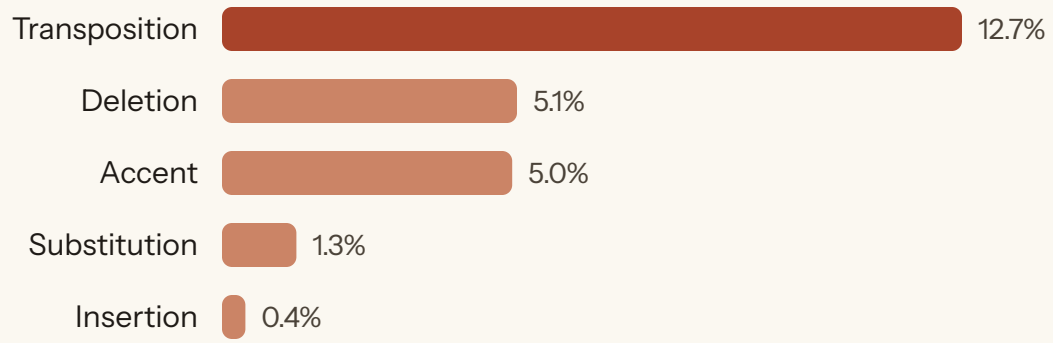


IBGE list of municipalities (localities API), downloaded Oct 21, 2026: 5,571 municipalities, 5,298 distinct names. 5,571 queries per number of errors.

ERROR KINDS

With 1 error, a swap pushes the right name out of first place in 12.7% of queries, 2.5x a deletion

% of 1-error queries where the right name lost first place, by error kind



transposition: 1,048 queries; deletion: 1,523 queries; accent: 461 queries; substitution: 1,291 queries; insertion: 1,248 queries.

HOW TO READ

- **Longer bar**: the kind of error that most often knocks the right name out of first.
- **Transposition**: swapping two neighboring letters counts as 2 Levenshtein edits.
- **Accent**: on lowercase text, each missing accented letter counts as 1 edit.

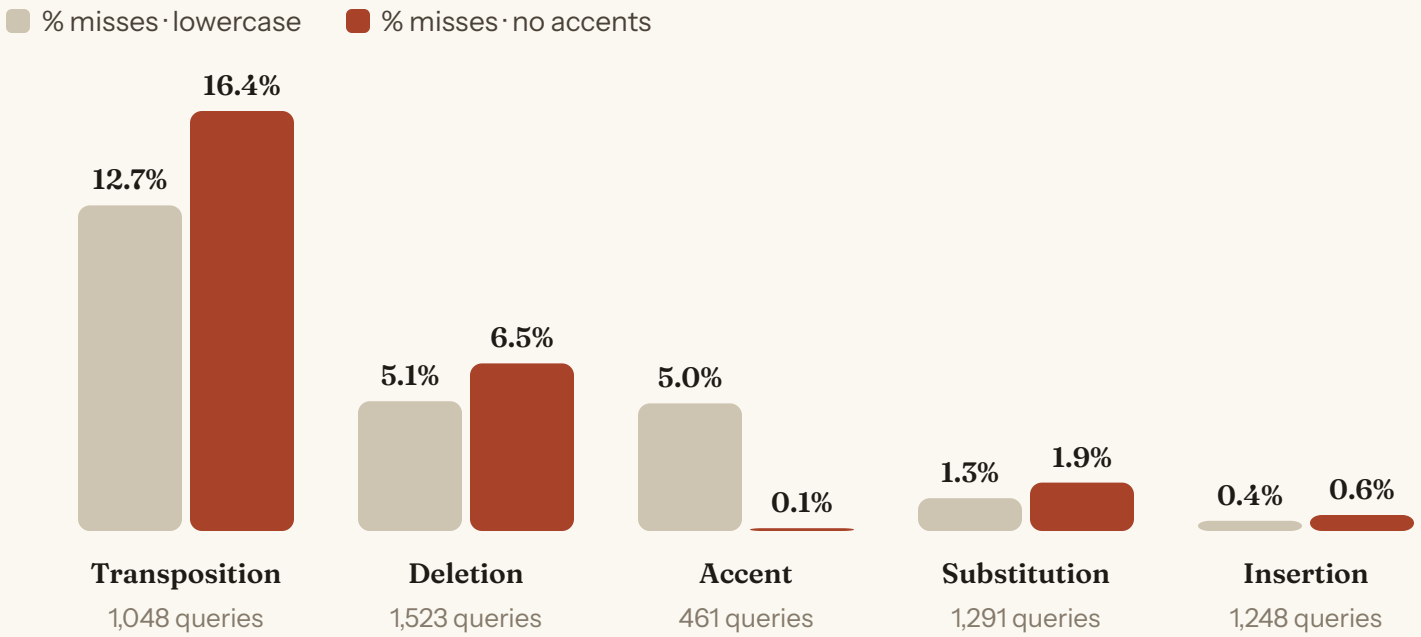
HYPOTHESIS TO TEST

The Damerau variant, which counts a swap as 1 edit, should reduce these misses. It is a hypothesis: it was not measured here.

IBGE list of municipalities (localities API), downloaded Oct 21, 2026: 5,571 municipalities, 5,298 distinct names. 5,571 queries with 1 error, lowercase text.

REMOVING ACCENTS

Removing accents fixes accent errors but raises misses for the other kinds: 85.0% against 85.5% overall



WHAT CHANGED

Without accents, accent errors almost vanish (5.0% → 0.1%). For the other kinds, misses go up: for swaps, from 12.7% to 16.4%.

HYPOTHESIS

When people type the accent, it helps separate similar names; removing it brings more names close to the query. For: misses rise for every kind except accents. Not measured: how many extra ties appear.

IBGE list of municipalities (localities API), downloaded Oct 21, 2026: 5,571 municipalities, 5,298 distinct names. Queries with 1 error; no accents also means no hyphens and apostrophes.

VERDICT

Strengths and limits

STRENGTHS

- Ranks the right name first in 85.5% of queries with 1 to 3 errors; exact matching gets 0.4%.
- With 1 error, first in 95.4% and in the top 5 in 99.8%.
- Names of 13 letters or more almost never miss (up to 0.9% misses).

LIMITS

- Names of up to 6 letters are 40% of misses, with 14% of queries.
- Swapping neighboring letters counts as 2 edits: 12.7% misses with 1 error.
- By name alone, it cannot tell apart municipalities that share a name (9.1% of them).

> Verdict: **good for suggesting** a list of municipalities; for correcting the query without confirmation, the first-place rate (85.5%) leaves about 1 query wrong in every 7.

IBGE list of municipalities (localities API), downloaded Oct 21, 2026: 5,571 municipalities, 5,298 distinct names. 16,713 queries with 1 to 3 errors, lowercase text.

HOW TO USE IT TODAY

Use the search to suggest a list of names, not to correct the query on its own

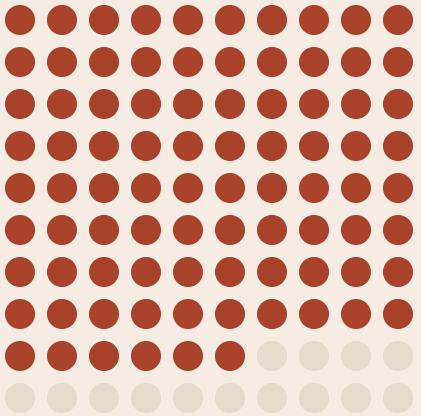
Use for

- show the 5 nearest names as suggestions: the right one is among them in 94.9% of queries

Don't use for

- replace the query with the first name without confirmation, or pick between same-name municipalities without the state

WHY



Out of 100 queries with typos, **86** bring the right name first. In a list of 5 suggestions, **95**.

IBGE list of municipalities (localities API), downloaded Oct 21, 2026: 5,571 municipalities, 5,298 distinct names. 16,713 queries with 1 to 3 errors, lowercase text.

NEXT STEPS

Proposed next steps

Count a swap of letters as 1 edit (Damerau)

A swap pushes the right name out of first place in 12.7% of 1-error queries.

Condition: the gain being measured on these same queries

Remove accents only when the query has none

Keeps the gain on accent errors without losing what a typed accent tells.

Condition: the overall result beating lowercase on the same queries

Ask for or infer the state for shared names

9.1% of municipalities share their name with another one in another state.

Condition: the search having the state or the person's location

Show 5 suggestions instead of correcting

The right name is in the top 5 in 94.9% of queries with typos.

Condition: the interface letting people choose

IBGE list of municipalities (localities API), downloaded Oct 21, 2026: 5,571 municipalities, 5,298 distinct names. 16,713 queries with 1 to 3 errors, lowercase text.

Fuzzy search over municipality names

A cutroom-deck example, made with cutroom.

Data: IBGE's list of municipalities. Code and numbers: examples/deck-fuzzy-search/src.

paper. github.com/galvaowesley/cutroom